

BTCS601

VI Semester Examination, May – June 2022

B.Tech. (CSE-ICS)

CONCEPT OF SYSTEM SECURITY

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

Minimum Pass Marks: 24

- Note:
1. All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 2. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 3. Draw neat diagram, wherever necessary.
 4. Assume suitable data wherever necessary.

Q.1(A) Define System Security Goals. 03

(B) What are the three pillars of security CIA? 03

(C) Explain characteristics of system security with principle of easiest penetration.

OR

(D) What do you understand by Risk, Vulnerability & Threat in a network? 06

Q.2(A) What are the Security vulnerability detection tools and techniques? 03

(B) How vulnerabilities in network can be identified? Explain. 03

(C) Define multics reference models for vulnerability analysis.

OR

(D) Explain hardware vulnerability and software vulnerability with example. 06

Q.3(A) What are the Operating System's Security Goals? 03

(B) Write about Threat Model with suitable example. 03

(C) Explain Lampson's Access Matrix.

OR

(D) Explain Mandatory protection system. 06

Contd...

- Q.4(A)** Define UNIX security goals and enlist its features. **03**
(B) Elaborate information flow integrity model. **03**
(C) Explain different Security models in Operating System.

OR

- (D)** How security bugs in windows Operating System can be traced? Explain. **06**

- Q.5(A)** What do you mean by Rooting Android devices? **03**
(B) Write about Attacks on Apps. **03**
(C) Explain Access control mechanism in Android operating system.

OR

- (D)** Enlist characteristics of Smartphone security. **06**
